Lecture 3

Solvability and Lie's theorem

Objectives. *Engel's theorem produced a common null vector. We now look for a common eigenvector. For a solvable algebra, every nonzero module has one, and every module can then be put in upper triangular form. Such a module need not be a direct sum of irreducible submodules. Solvable ideals also lead to the radical. Removing the radical leaves a semisimple algebra, the subject of the next lectures.*

By the end of this lecture you will be able to:

- Use derived series to distinguish solvability from nilpotency.
- Prove that the radical exists.
- Reconstruct the invariance lemma and its trace step.
- Construct an invariant flag in a module of a solvable algebra.

Throughout, Lie algebras and modules are finite-dimensional over $\mathbb{C}$.

3.1 The derived series

Definition 3.1 (Derived series and solvability)**.** The *derived series* of $\mathfrak{g}$ is defined by

$$\mathfrak{g}^{(0)} = \mathfrak{g}, \qquad \mathfrak{g}^{(k+1)} = [\mathfrak{g}^{(k)}, \mathfrak{g}^{(k)}].$$

The algebra $\mathfrak{g}$ is *solvable* if $\mathfrak{g}^{(m)} = 0$ for some $m \geq 0$.

Each $\mathfrak{g}^{(k)}$ is an ideal of $\mathfrak{g}$. To see this, let I and J be ideals, and take $x \in \mathfrak{g}$, $i \in I$ and $j \in J$. Jacobi gives

$$[x, [i, j]] = [[x, i], j] + [i, [x, j]] \in [I, J].$$

So $[I, J]$ is an ideal. Induction on k finishes the argument.

An abelian algebra $\mathfrak{g}$ is solvable, since $\mathfrak{g}^{(1)} = 0$. The basic nonabelian example is the two-dimensional algebra

$$\mathfrak{g} = \mathbb{C}h \oplus \mathbb{C}e, \qquad [h, e] = e. \tag{3.1}$$

Its derived series is

$$\mathfrak{g}^{(1)} = \mathbb{C}e, \qquad \mathfrak{g}^{(2)} = [\mathbb{C}e, \mathbb{C}e] = 0.$$

Lemma 3.2 (Closure properties)**.** *(a) If $\mathfrak{s}$ is a subalgebra of $\mathfrak{g}$, then $\mathfrak{s}^{(k)} \subseteq \mathfrak{g}^{(k)}$ for all k.*
(b) If $\varphi : \mathfrak{g} \to \mathfrak{l}$ is a surjective homomorphism, then $\varphi(\mathfrak{g}^{(k)}) = \mathfrak{l}^{(k)}$ for all k.
(c) Let $\mathfrak{a}$ be an ideal of $\mathfrak{g}$. If $\mathfrak{a}$ and $\mathfrak{g}/\mathfrak{a}$ are solvable, then $\mathfrak{g}$ is solvable.
In particular, subalgebras and quotients of solvable algebras are solvable.

Proof. Part (a) follows by induction, since $[\mathfrak{s}^{(k)}, \mathfrak{s}^{(k)}] \subseteq [\mathfrak{g}^{(k)}, \mathfrak{g}^{(k)}]$. For (b), a homomorphism satisfies

$$\varphi([U, W]) = [\varphi(U), \varphi(W)]$$

for subspaces U and W of $\mathfrak{g}$. Induction from $\varphi(\mathfrak{g}) = \mathfrak{l}$ gives the claim.

For (c), choose m and n with $(\mathfrak{g}/\mathfrak{a})^{(m)} = 0$ and $\mathfrak{a}^{(n)} = 0$. Apply (b) to the quotient map $q\colon \mathfrak{g} \to \mathfrak{g}/\mathfrak{a}$:

$$q(\mathfrak{g}^{(m)}) = (\mathfrak{g}/\mathfrak{a})^{(m)} = 0, \qquad \text{so} \qquad \mathfrak{g}^{(m)} \subseteq \mathfrak{a}.$$

The derived series of $\mathfrak{g}^{(m)}$ is $\mathfrak{g}^{(m)}, \mathfrak{g}^{(m+1)}, \dots$. Part (a), applied to the subalgebra $\mathfrak{g}^{(m)}$ of $\mathfrak{a}$, gives

$$\mathfrak{g}^{(m+n)} = (\mathfrak{g}^{(m)})^{(n)} \subseteq \mathfrak{a}^{(n)} = 0. \qquad \blacksquare$$

We index the lower central series of Lecture 2 by

$$C^0\mathfrak{g} = \mathfrak{g}, \qquad C^{k+1}\mathfrak{g} = [\mathfrak{g}, C^k\mathfrak{g}].$$

Thus $\mathfrak{g}$ is nilpotent when $C^m\mathfrak{g} = 0$ for some m.

Proposition 3.3 (Nilpotent versus solvable)**.** *Every nilpotent Lie algebra is solvable. The algebra* (3.1) *is solvable but not nilpotent.*

Proof. We show $\mathfrak{g}^{(k)} \subseteq C^k\mathfrak{g}$ by induction. The case $k = 0$ is an equality. If the inclusion holds for k, then

$$\mathfrak{g}^{(k+1)} = [\mathfrak{g}^{(k)}, \mathfrak{g}^{(k)}] \subseteq [\mathfrak{g}, C^k\mathfrak{g}] = C^{k+1}\mathfrak{g}.$$

Thus $C^m\mathfrak{g} = 0$ forces $\mathfrak{g}^{(m)} = 0$.

For the algebra (3.1), $C^1\mathfrak{g} = \mathbb{C}e$. If $C^k\mathfrak{g} = \mathbb{C}e$ with $k \geq 1$, then

$$C^{k+1}\mathfrak{g} = [\mathfrak{g}, \mathbb{C}e] = \mathbb{C}[h, e] = \mathbb{C}e.$$

No term of the lower central series is zero. $\blacksquare$

In the algebra (3.1), the ideal $\mathbb{C}e$ and the quotient $\mathfrak{g}/\mathbb{C}e$ are abelian, hence nilpotent. The algebra $\mathfrak{g}$ is not nilpotent. Thus the assertion about ideals and quotients in Lemma 3.2 has no nilpotent analogue.

3.2 The radical

Proposition 3.4 (Sums of solvable ideals)**.** *If $\mathfrak{a}$ and $\mathfrak{c}$ are solvable ideals of $\mathfrak{g}$, then $\mathfrak{a} + \mathfrak{c}$ is a solvable ideal of $\mathfrak{g}$.*

Proof. A sum of ideals is an ideal. Restricting the quotient map of $\mathfrak{a} + \mathfrak{c}$ modulo its ideal $\mathfrak{c}$ to $\mathfrak{a}$ gives the homomorphism

$$\mathfrak{a} \longrightarrow (\mathfrak{a} + \mathfrak{c})/\mathfrak{c}, \qquad x \longmapsto x + \mathfrak{c}.$$

This homomorphism is surjective, because $(x + y) + \mathfrak{c} = x + \mathfrak{c}$ for $y \in \mathfrak{c}$.

Since $\mathfrak{a}$ is solvable, Lemma 3.2 for homomorphic images makes the quotient $(\mathfrak{a} + \mathfrak{c})/\mathfrak{c}$ solvable. The ideal $\mathfrak{c}$ is solvable, so Lemma 3.2 for an ideal and the quotient by it makes $\mathfrak{a} + \mathfrak{c}$ solvable. $\blacksquare$

Definition 3.5 (Radical and semisimplicity)**.** The *radical* $\operatorname{rad}\mathfrak{g}$ is a solvable ideal of $\mathfrak{g}$ that contains every solvable ideal of $\mathfrak{g}$. The algebra $\mathfrak{g}$ is *semisimple* if $\operatorname{rad}\mathfrak{g} = 0$.

Proposition 3.6 (Existence of the radical)**.** *Every $\mathfrak{g}$ has a unique radical $\mathfrak{r}$. Moreover,* $\operatorname{rad}(\mathfrak{g}/\mathfrak{r}) = 0$.

Proof. The zero ideal is solvable. Choose a solvable ideal $\mathfrak{r}$ of maximal dimension. For a solvable ideal $\mathfrak{a}$, Proposition 3.4 makes $\mathfrak{r} + \mathfrak{a}$ a solvable ideal. Maximality gives

$$\mathfrak{r} + \mathfrak{a} = \mathfrak{r}, \qquad \text{so} \qquad \mathfrak{a} \subseteq \mathfrak{r}.$$

Two radicals contain each other, so they are equal.

Now let $q\colon \mathfrak{g} \to \mathfrak{g}/\mathfrak{r}$ be the quotient map, and let $\bar{\mathfrak{a}}$ be a solvable ideal of $\mathfrak{g}/\mathfrak{r}$. The preimage $\mathfrak{a} = q^{-1}(\bar{\mathfrak{a}})$ is an ideal of $\mathfrak{g}$ containing $\mathfrak{r}$, and $\mathfrak{a}/\mathfrak{r} \cong \bar{\mathfrak{a}}$ is solvable.

Lemma 3.2 for an ideal and the quotient by it makes $\mathfrak{a}$ solvable. Hence $\mathfrak{a} \subseteq \mathfrak{r}$, and $\bar{\mathfrak{a}} = 0$. ∎

Solvable ideals are the part removed by the radical. Semisimple algebras have no nonzero solvable ideal. Proposition 3.6 says that the quotient $\mathfrak{g}/\operatorname{rad}\mathfrak{g}$ is semisimple. From Lecture 4 on, we study semisimple algebras and their representations.

A solvable algebra is its own radical. For $\mathfrak{sl}_2$, Lecture 1 gives $[\mathfrak{sl}_2, \mathfrak{sl}_2] = \mathfrak{sl}_2$, so every term of its derived series is $\mathfrak{sl}_2$. Hence $\mathfrak{sl}_2$ is not solvable.

Let $I \neq 0$ be an ideal of $\mathfrak{sl}_2$. It is stable under $\operatorname{ad} h$, whose distinct eigenspaces are $\mathbb{C}h$, $\mathbb{C}e$ and $\mathbb{C}f$. By Lagrange interpolation, the projections onto these eigenspaces are polynomials in $\operatorname{ad} h$. Hence I contains at least one of h, e and f. The three bracket relations of Lecture 1 then force I to contain all three. Thus the only ideals are 0 and $\mathfrak{sl}_2$. Since $\mathfrak{sl}_2$ is not solvable, its radical is zero, so it is semisimple.

A solvable and semisimple algebra equals its radical, which is zero. So only the zero algebra has both properties. An algebra can also have neither. Exercise 3.4 proves that

$$\operatorname{rad}\mathfrak{gl}_2(\mathbb{C}) = \mathbb{C}I_2, \qquad \mathfrak{gl}_2(\mathbb{C})/\mathbb{C}I_2 \cong \mathfrak{sl}_2.$$

The radical of $\mathfrak{gl}_2(\mathbb{C})$ is neither 0 nor $\mathfrak{gl}_2(\mathbb{C})$, so this algebra is neither semisimple nor solvable.

Figure 3.1 places the examples so far. The upper triangular algebra $\mathfrak{b} \subseteq \mathfrak{gl}_2(\mathbb{C})$ of Lecture 1 is solvable, since $[\mathfrak{b}, \mathfrak{b}] = \mathbb{C}e$ is abelian. Lecture 2 shows that it is not nilpotent.

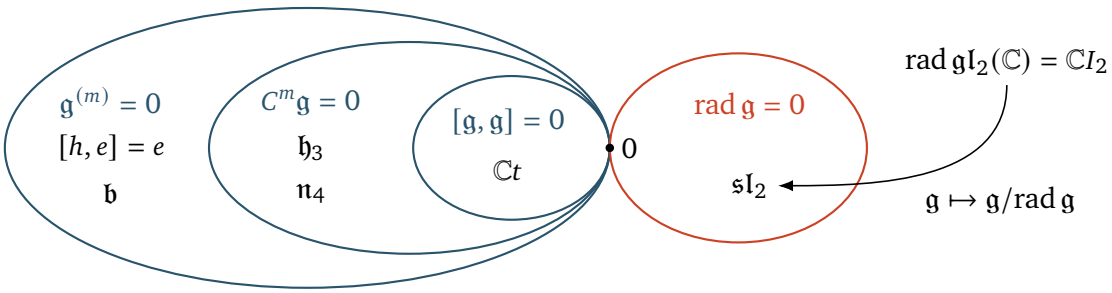


Figure 3.1: The first examples of Lie algebras, each region labelled by its defining condition for some m: abelian inside nilpotent inside solvable, and on the right semisimple, which meets the others only in the zero algebra. The algebra $\mathfrak{gl}_2(\mathbb{C})$ lies outside every region, and the arrow is its quotient by the radical.

3.3 Lie's theorem

Let $\mathfrak{k}$ be a subalgebra of $\mathfrak{g}$, let V be a $\mathfrak{g}$-module and let $\lambda : \mathfrak{k} \to \mathbb{C}$ be linear. Define the subspace

$$V_\lambda^{\mathfrak{k}} = \{v \in V : \rho(y)v = \lambda(y)v \text{ for all } y \in \mathfrak{k}\}.$$

A nonzero vector of $V_\lambda^{\mathfrak{k}}$ is a *common eigenvector* of $\mathfrak{k}$ in V. The map λ is its *eigenvalue function*.

The proof of Lie's theorem extends a common eigenvector from an ideal to the whole algebra.

Lemma 3.7 (Invariance lemma). *Let $\mathfrak{k}$ be an ideal of $\mathfrak{g}$, let V be a $\mathfrak{g}$-module and let $\lambda : \mathfrak{k} \to \mathbb{C}$ be linear. Then*

$$\rho(x)V_\lambda^{\mathfrak{k}} \subseteq V_\lambda^{\mathfrak{k}} \quad \text{for all } x \in \mathfrak{g}.$$

Proof. Write $W = V_\lambda^{\mathfrak{k}}$. We may assume $W \neq 0$. Fix $x \in \mathfrak{g}$.

The lemma rests on a commutator identity. For $v \in W$ and $y \in \mathfrak{k}$, the representation property gives

$$\rho(y)\rho(x)v = \rho(x)\rho(y)v - \rho([x,y])v = \lambda(y)\rho(x)v - \lambda([x,y])v.$$

The last step uses $[x,y] \in \mathfrak{k}$, which holds since $\mathfrak{k}$ is an ideal. So $\rho(x)v \in W$ as soon as $\lambda([x,y]) = 0$ for all $y \in \mathfrak{k}$. We force this last scalar to vanish by taking the trace of a commutator on a subspace U preserved by $\rho(x)$ and by every $\rho(y)$. Steps 1 to 3 construct U and compute this trace. Step 4 concludes.

Now fix a nonzero $w \in W$.

Step 1: a cyclic subspace. Put $U_0 = 0$ and

$$U_j = \operatorname{span}\{w, \rho(x)w, \dots, \rho(x)^{j-1}w\} \qquad (j \geq 1).$$

Let n be the least integer with $\rho(x)^n w \in U_n$. Such an integer exists because the ascending chain U_j stabilises in the finite-dimensional space V. Since $w \neq 0$, we have $n \geq 1$.

The vectors $w, \dots, \rho(x)^{n-1}w$ are independent. In a dependence, the last nonzero coefficient, at some index $j < n$, would give $\rho(x)^j w \in U_j$, contrary to the minimality of n.

Put $U = U_n$. Then $\dim U = n$. The defining relation $\rho(x)^n w \in U$ gives $\rho(x)U \subseteq U$.

Step 2: the triangular action of $\mathfrak{k}$. We prove, by induction on $j \geq 0$, that

$$\rho(y)\rho(x)^j w \in \lambda(y)\rho(x)^j w + U_j \quad \text{for all } y \in \mathfrak{k}. \tag{3.2}$$

For $j = 0$, this is $\rho(y)w = \lambda(y)w$. For $j \geq 1$, put $u = \rho(x)^{j-1}w$. The representation property gives

$$\rho(y)\rho(x)u = \rho(x)\rho(y)u - \rho([x,y])u.$$

Since $\mathfrak{k}$ is an ideal, $[x,y] \in \mathfrak{k}$. The induction hypothesis for y and for $[x,y]$ gives

$$\rho(y)u \in \lambda(y)u + U_{j-1}, \qquad \rho([x,y])u \in \lambda([x,y])u + U_{j-1} \subseteq U_j.$$

Apply $\rho(x)$ to the first inclusion. Since $\rho(x)U_{j-1} \subseteq U_j$, we obtain

$$\rho(x)\rho(y)u \in \lambda(y)\rho(x)^j w + U_j.$$

Substituting the two inclusions into the preceding identity gives (3.2).

Step 3: the trace. Use the basis

$$\big(w,\ \rho(x)w,\ \dots,\ \rho(x)^{n-1}w\big)$$

of U. By (3.2), each $\rho(y)$ with $y \in \mathfrak{k}$ preserves U. Its matrix in this basis is

$$\begin{pmatrix} \lambda(y) & * & \cdots & * \\ 0 & \lambda(y) & \cdots & * \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda(y) \end{pmatrix}.$$

Hence

$$\operatorname{tr}\big(\rho(y)|_U\big) = n\,\lambda(y) \qquad (y \in \mathfrak{k}).$$

Apply this to $[x, y] \in \mathfrak{k}$. Both $\rho(x)$ and $\rho(y)$ preserve U, so

$$\begin{aligned} n\,\lambda([x,y]) &= \operatorname{tr}\big(\rho([x,y])|_U\big) \\ &= \operatorname{tr}\big(\rho(x)|_U\,\rho(y)|_U - \rho(y)|_U\,\rho(x)|_U\big) \\ &= 0. \end{aligned}$$

Since $n \geq 1$ and $\mathbb{C}$ has characteristic zero, $\lambda([x, y]) = 0$ for all $y \in \mathfrak{k}$.

Step 4: invariance. Step 3 gives $\lambda([x, y]) = 0$, independently of w. For every $v \in W$ and $y \in \mathfrak{k}$,

$$\begin{aligned} \rho(y)\rho(x)v &= \rho(x)\rho(y)v - \rho([x,y])v \\ &= \lambda(y)\rho(x)v - \lambda([x,y])v \\ &= \lambda(y)\rho(x)v. \end{aligned}$$

Thus $\rho(x)v \in W$. ∎

Theorem 3.8 (Lie's theorem). *Let $\mathfrak{g}$ be solvable and let $V \neq 0$ be a $\mathfrak{g}$-module. There are a nonzero $v \in V$ and a linear map $\lambda : \mathfrak{g} \to \mathbb{C}$ such that*

$$\rho(x)v = \lambda(x)v \quad \textit{for all } x \in \mathfrak{g}.$$

Every such λ vanishes on $[\mathfrak{g}, \mathfrak{g}]$.

Proof. We first prove the last statement, which holds for any $\mathfrak{g}$. For $x, y \in \mathfrak{g}$,

$$\lambda([x,y])v = \rho(x)\rho(y)v - \rho(y)\rho(x)v = \big(\lambda(x)\lambda(y) - \lambda(y)\lambda(x)\big)v = 0.$$

Since $v \neq 0$, this gives $\lambda([x, y]) = 0$.

Existence is proved by induction on $\dim \mathfrak{g}$. If $\mathfrak{g} = 0$, take any nonzero v and $\lambda = 0$.

Let $\dim \mathfrak{g} \geq 1$. If $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$, then every $\mathfrak{g}^{(k)}$ would equal $\mathfrak{g} \neq 0$, and $\mathfrak{g}$ would not be solvable. Hence $[\mathfrak{g}, \mathfrak{g}] \neq \mathfrak{g}$. Choose a subspace $\mathfrak{k}$ of codimension one with $[\mathfrak{g}, \mathfrak{g}] \subseteq \mathfrak{k}$. The subspace $\mathfrak{k}$ is an ideal, because

$$[\mathfrak{g}, \mathfrak{k}] \subseteq [\mathfrak{g}, \mathfrak{g}] \subseteq \mathfrak{k}.$$

It is solvable by Lemma 3.2 for subalgebras. By induction, $\mathfrak{k}$ has a common eigenvector in V, with some eigenvalue function λ. Thus $W = V_\lambda^{\mathfrak{k}} \neq 0$.

Choose $z \in \mathfrak{g} \setminus \mathfrak{k}$, so that $\mathfrak{g} = \mathfrak{k} \oplus \mathbb{C}z$. By Lemma 3.7, $\rho(z)W \subseteq W$. Since $\mathbb{C}$ is algebraically closed, $\rho(z)|_W$ has an eigenvector v, say $\rho(z)v = cv$. Define $\tilde{\lambda} : \mathfrak{g} \to \mathbb{C}$ by

$$\tilde{\lambda}(y + tz) = \lambda(y) + tc \qquad (y \in \mathfrak{k},\ t \in \mathbb{C}).$$

Then

$$\rho(y + tz)v = \lambda(y)v + tcv = \tilde{\lambda}(y + tz)v.$$ ∎

The proof used characteristic zero once, to divide by n in Step 3 of the invariance lemma. It used algebraic closure once, for the eigenvector of $\rho(z)$. Exercise 3.6 shows that the theorem fails if either hypothesis is dropped.

An optional remark in Lecture 12 returns to this theorem. It applies the theorem to a solvable subalgebra of a semisimple algebra. This gives another way to find the vector that Lecture 12 uses to generate a finite-dimensional irreducible module. Nothing in this lecture depends on that remark.

3.4 Triangular form

Corollary 3.9 (Invariant flags). *Let $\mathfrak{g}$ be solvable and let V be a $\mathfrak{g}$-module of dimension d. There is a basis $(v_1, \dots, v_d)$ of V such that every*

$$V_i = \operatorname{span}\{v_1, \dots, v_i\} \qquad (0 \le i \le d)$$

is a submodule. In this basis the matrix $D(x)$ of every $\rho(x)$ is upper triangular. Its diagonal entries are $D(x)_{ii} = \lambda_i(x)$, where each $\lambda_i \colon \mathfrak{g} \to \mathbb{C}$ is a linear map that vanishes on $[\mathfrak{g}, \mathfrak{g}]$.

Proof. We induct on d. The case $d = 0$ is empty. For $d \ge 1$, Lie's theorem gives a common eigenvector v_1, so $V_1 = \mathbb{C}v_1$ is a submodule.

By induction, the module V/V_1 has a basis $(\bar{u}_2, \dots, \bar{u}_d)$ whose initial spans are submodules. Choose $v_i \in V$ with $v_i + V_1 = \bar{u}_i$. Since the induced action of x maps $\bar{u}_i$ into $\operatorname{span}\{\bar{u}_2, \dots, \bar{u}_i\}$,

$$\rho(x)v_i \in V_1 + \operatorname{span}\{v_2, \dots, v_i\} = V_i.$$

This gives the invariant flag

$$0 = V_0 \subset V_1 \subset \cdots \subset V_d = V,$$

and each $D(x)$ is upper triangular.

The entry $D(x)_{ii}$ is the scalar by which x acts on the one-dimensional module V_i/V_{i-1}. Theorem 3.8, applied to that module, shows that λ_i vanishes on $[\mathfrak{g}, \mathfrak{g}]$. ∎

Corollary 3.10 (Irreducible modules). *Let $\mathfrak{g}$ be solvable. Every irreducible $\mathfrak{g}$-module is one-dimensional. For linear $\lambda \colon \mathfrak{g} \to \mathbb{C}$ with $\lambda([\mathfrak{g}, \mathfrak{g}]) = 0$, let $\mathbb{C}_\lambda$ be $\mathbb{C}$ with the action*

$$\rho_\lambda(x)c = \lambda(x)c \qquad (x \in \mathfrak{g},\ c \in \mathbb{C}).$$

Then $\lambda \mapsto \mathbb{C}_\lambda$ is a bijection from these maps to the isomorphism classes of irreducible modules.

Proof. Let V be an irreducible module. By Lie's theorem, V has a common eigenvector v, with some eigenvalue function λ. The line $\mathbb{C}v$ is a nonzero submodule, so $\mathbb{C}v = V$. Theorem 3.8 shows that λ vanishes on $[\mathfrak{g}, \mathfrak{g}]$, and $c \mapsto cv$ is an isomorphism $\mathbb{C}_\lambda \to V$.

Conversely, let $\lambda \colon \mathfrak{g} \to \mathbb{C}$ be linear and vanish on $[\mathfrak{g}, \mathfrak{g}]$. Then

$$\lambda([x, y]) = 0 = \lambda(x)\lambda(y) - \lambda(y)\lambda(x),$$

so $\mathbb{C}_\lambda$ is a module. It is irreducible because it is one-dimensional.

An isomorphism $\mathbb{C}_\lambda \to \mathbb{C}_\mu$ is multiplication by some $t \neq 0$. The intertwining condition $t\lambda(x) = \mu(x)t$ forces $\lambda = \mu$. ∎

We now characterise representations that admit upper triangular matrices. Let $\mathfrak{b}_d$ be the algebra of upper triangular $d \times d$ matrices, and $\mathfrak{n}_d$ its subalgebra of strictly upper triangular matrices.

Theorem 3.11 (Triangular representations). *Let $\mathfrak{g}$ be any Lie algebra and let $\rho\colon \mathfrak{g} \to \mathfrak{gl}(V)$ be a representation with $\dim V = d$. Some basis of V makes the matrix $D(x)$ of every $\rho(x)$ upper triangular if and only if $\rho(\mathfrak{g})$ is solvable. If ρ is faithful, this holds if and only if $\mathfrak{g}$ is solvable.*

Proof. If $\rho(\mathfrak{g})$ is solvable, then V is a module for the solvable algebra $\rho(\mathfrak{g}) \subseteq \mathfrak{gl}(V)$. Corollary 3.9 gives a basis that triangularises every $\rho(x)$.

Conversely, such a basis identifies $\rho(\mathfrak{g})$ with a subalgebra of $\mathfrak{b}_d$. For $X, Y \in \mathfrak{b}_d$,

$$(XY)_{ii} = X_{ii}Y_{ii} = (YX)_{ii},$$

so $[X, Y] \in \mathfrak{n}_d$. Thus $\mathfrak{b}_d^{(1)} \subseteq \mathfrak{n}_d$.

Lecture 2 proved that $\mathfrak{n}_d$ is nilpotent. By Proposition 3.3, $\mathfrak{n}_d^{(m)} = 0$ for some m. Lemma 3.2 for subalgebras gives

$$\mathfrak{b}_d^{(m+1)} = (\mathfrak{b}_d^{(1)})^{(m)} \subseteq \mathfrak{n}_d^{(m)} = 0.$$

The same lemma makes the subalgebra $\rho(\mathfrak{g})$ solvable. If ρ is faithful, then $\mathfrak{g} \cong \rho(\mathfrak{g})$. ∎

Faithfulness matters. The zero representation of $\mathfrak{sl}_2$ is triangular, but $\mathfrak{sl}_2$ is not solvable.

Remark 3.12 (Linear Jordan decomposition). Recall from linear algebra (Humphreys, §4.2) that each $T \in \mathrm{End}(V)$ has a unique decomposition

$$\begin{aligned} T &= T_s + T_n, \\ T_s &\text{ is diagonalisable,} \\ T_n &\text{ is nilpotent,} \\ T_sT_n &= T_nT_s. \end{aligned}$$

Both parts are polynomials in T without constant term.

In the basis of Corollary 3.9, the diagonal entries of $D(x)$ are the eigenvalues of $\rho(x)$, with multiplicity. These are also the eigenvalues of $\rho(x)_s$. The diagonal part of $D(x)$ need not be the matrix of $\rho(x)_s$. The parts $\rho(x)_s$ and $\rho(x)_n$ need not lie in $\rho(\mathfrak{g})$. Exercise 3.7 gives examples of both. For semisimple $\mathfrak{g}$, Lecture 4 cites a Jordan decomposition inside $\mathfrak{g}$ itself, and Lecture 7 uses it.

3.5 A line without an invariant complement

Example 3.13 (The algebra $[h, e] = e$ on $\mathbb{C}^2$). Let $\mathfrak{g}$ be the algebra (3.1). Let $\rho(h)$ and $\rho(e)$ be the operators on $\mathbb{C}^2$ whose matrices in the standard basis (v_1, v_2) are

$$D(h) = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad D(e) = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

The products are

$$D(h)D(e) = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad D(e)D(h) = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix},$$

so $[D(h), D(e)] = D(e)$. This is a representation. It is faithful, since $D(ah + be)$ has first row (a, b).

Following the proof. Here $[\mathfrak{g},\mathfrak{g}] = \mathbb{C}e$, so take $\mathfrak{k} = \mathbb{C}e$. The matrix $D(e)$ is nilpotent, so its only eigenvalue is 0 and $\lambda(e) = 0$. Then

$$W = V_0^{\mathfrak{k}} = \ker D(e) = \mathbb{C}v_1.$$

The invariance lemma predicts $D(h)W \subseteq W$. Indeed $D(h)v_1 = v_1$.

For $x = h$ and $w = v_1$, the cyclic subspace is $U = \mathbb{C}v_1$, with $n = 1$. Step 3 of the invariance lemma reads

$$1 \cdot \lambda([h,e]) = \lambda(e) = \operatorname{tr}\big(\rho(e)|_U\big) = 0.$$

The final step uses the eigenvector v_1 of $\rho(h)$, with eigenvalue 1. The resulting eigenvalue function is

$$\lambda_1(ah + be) = a,$$

which vanishes on $[\mathfrak{g},\mathfrak{g}] = \mathbb{C}e$.

The flag. The flag $0 \subset \mathbb{C}v_1 \subset \mathbb{C}^2$ is invariant. On the quotient $\mathbb{C}^2/\mathbb{C}v_1$, spanned by $v_2 + \mathbb{C}v_1$,

$$\rho(h)v_2 = 0, \qquad \rho(e)v_2 = v_1 \in \mathbb{C}v_1.$$

So the quotient is the trivial module $\mathbb{C}_0$, and $\lambda_2 = 0$.

No invariant complement. An invariant line is spanned by an eigenvector of $D(e)$. The only eigenvalue of $D(e)$ is zero, and $\ker D(e) = \mathbb{C}v_1$. Hence $\mathbb{C}v_1$ is the only invariant line, and it has no invariant complement.

Every irreducible $\mathfrak{g}$-module is one-dimensional, by Corollary 3.10. Since $\mathbb{C}^2$ has only one invariant line, it is not a direct sum of irreducible submodules. Lecture 5 proves that every finite-dimensional module of a complex semisimple algebra is such a direct sum.

We close with an outlook. It is not used later. Over a field of characteristic zero, the radical exists by the same argument.

Theorem 3.14 (Levi decomposition · *C*)**.** *Let $\mathfrak{g}$ be a finite-dimensional Lie algebra over a field of characteristic zero. There is a subalgebra $\mathfrak{s}$ of $\mathfrak{g}$ with*

$$\mathfrak{g} = \mathfrak{s} \oplus \operatorname{rad}\mathfrak{g}$$

as vector spaces. Source: Fulton–Harris, Representation Theory*, Appendix E, §E.1; Jacobson,* Lie Algebras*, Chapter III, §9.*

The quotient map restricts to an isomorphism $\mathfrak{s} \cong \mathfrak{g}/\operatorname{rad}\mathfrak{g}$. This quotient is semisimple by Proposition 3.6. For (3.1), $\operatorname{rad}\mathfrak{g} = \mathfrak{g}$ and $\mathfrak{s} = 0$.

Self-checks

(1) Can $\mathfrak{sl}_2$ have a faithful module in which every $D(x)$ is upper triangular?

(2) Let $\mathfrak{g}$ be solvable and V irreducible. How does $[\mathfrak{g},\mathfrak{g}]$ act on V?

(3) In Example 3.13, is $\mathbb{C}v_2$ a submodule? What is the quotient module $\mathbb{C}^2/\mathbb{C}v_1$?

Answers. (1) No. Theorem 3.11 would make $\mathfrak{sl}_2$ solvable, but $[\mathfrak{sl}_2,\mathfrak{sl}_2] = \mathfrak{sl}_2$. (2) By zero. Corollary 3.10 gives $V \cong \mathbb{C}_\lambda$ with $\lambda([\mathfrak{g},\mathfrak{g}]) = 0$. (3) No, since $D(e)v_2 = v_1$. The quotient is the trivial module $\mathbb{C}_0$.

Summary

The derived series defines solvability. Solvability passes to subalgebras and quotients. If an ideal and the quotient by it are solvable, then so is the whole algebra. The analogous statement for nilpotency fails. Sums of solvable ideals are solvable, so the radical exists. The quotient by the radical is semisimple.

Lie's theorem gives a common eigenvector in every nonzero module of a solvable algebra. Its main step is the invariance lemma, whose trace step needs characteristic zero. A representation is triangularisable exactly when its image is solvable. A triangular module need not be a direct sum of irreducible submodules.

Reading

Humphreys, *Introduction to Lie Algebras and Representation Theory*, Sections 3.1, 4.1 and 4.2. Erdmann–Wildon, *Introduction to Lie Algebras*, Chapters 4–6. Levi's theorem: Fulton–Harris, Appendix E.1. Exercise Sheet 3 gives further calculations and proofs.